If the Binary number is $(A_3\ A_2\ A_1\ A_0)_2$

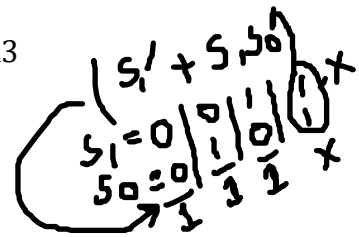
	S2	S1	S0	H3	H2	H1	H0
Logical-Left	0	0	0	**A2**	A1	A0	0
Circular -Left	0	0	1	**A2**	A1	A0	A3
Arithmetic-Left	0	1	0	**A3**	A1	A0	0
Logical-Right	1	0	0	0	A3	A2	**A1**
Circular -Right	1	0	1	A0	A3	A2	**A1**
Arithmetic-Right	1	1	0	A3	A3	A2	**A1**

H3 = S2S1'S0A0 + (S2'S1'S0'+S2'S1'S0)A2 + (S2'S1S0'+S2S1S0')A3

= S2S1'S0A0 + (S0'+ S0)S2'S1'A2 + (S2'+S2)S1S0'A3

= S2S1'S0A0 + S2'S1'A2 + S1S0'A3

H2 = (S2'S1'S0' + S2'S1'S0 + S2'S1S0')A1 + (S2S1'S0' + S2S1'S0 + S2S1S0')A3



= (S1'S0' + S1'S0 + S1S0')S2'A1 + (S1'S0' + S1'S0 + S1S0')S2A3

= (S1'S0' + S1'S0 + S1S0')(S2'A1 + S2A3)

= (S1'(S0'+S0) + S1S0') (S2'A1 + S2A3)

= (S1' + S1S0') (S2'A1 + S2A3)

= **S2'A1 + S2A3**

H1 = (S2'S1'S0' + S2'S1'S0 + S2'S1S0')A0 + (S2S1'S0' + S2S1'S0 + S2S1S0')A2

= (S1'S0' + S1'S0 + S1S0')S2'A0 + (S1'S0' + S1'S0 + S1S0')S2A2

= (S1'S0' + S1'S0 + S1S0')(S2'A0 + S2A2)

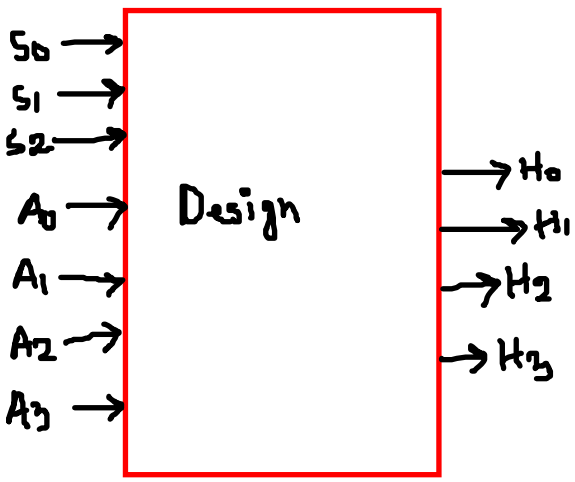
= (S1'(S0'+S0) + S1S0') (S2'A0 + S2A2)

= (S1' + S1S0') (S2'A0 + S2A2)

= **S2'A0 + S2A2**

H0 = S2'S1'S0A3 + S2A1

= **S2'S0A3 + S2A1**



4 bit Universal Shift Circuit